THE VIRTUAL NORMAL SHEAF CONSTRUCTION IS WELL DEFINED AS A BIVARIANT CLASS AND IS COMPATIBLE WITH BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FBK (sources/stacks-chow/pinned-original-source.tex line 11912).

Allowed standard prerequisites (editorial).

- Flat pullback, proper pushforward and the class C of the Gysin map at infinity as bivariant classes. Same chapter, Lemmas labelled lemma-flat-pullback-bivariant, lemma-push-proper-bivariant and lemma-gysin-at-infty, cited at proof lines 11926-11931 and 11960-11962. Makes both sides of the defining equation bivariant classes and supplies the class C used to define c.
- Bijectivity of the pullback along a vector bundle. Same chapter, Lemma labelled lemma-vectorbundle, cited at proof lines 11932-11934. Allows c to be defined from the equation by dividing by p^*.
- Base change of the virtual normal sheaf and the independence of C under pushforward. Same chapter, Lemma labelled lemma-pullback-virtual-normal-sheaf, cited at proof lines 11911-11915 and 11938-11940, and Lemma labelled lemma-gysin-at-infty-independent, cited at proof lines 11972-11976. Fixes the class N' of the base changed situation and compares the two classes C cap alpha' and C' cap alpha'.
- Restriction of bivariant classes and flat pullback of proper pushforwards. Same chapter, Remark labelled remark-restriction-bivariant, invoked in the statement, and Lemma labelled lemma-flat-pullback-proper-pushforward, cited at proof lines 11978-11982. Gives the meaning of the compatibility statement and transports the identity to the opens.

Source reference. Original reference 0FE0: sources/stacks-chow/pinned-original-source.tex, source lines 6675–6675. Same-chapter pointer into the author environment remark at source line 6675; invoked at proof line None of the counted proof.

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Source reference. Original lemma 0FBJ: sources/stacks-chow/pinned-original-source.tex, source lines 11851–11851. Same-chapter pointer into the author environment lemma at source line 11851; invoked at proof line 11939 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B78: sources/stacks-chow/pinned-original-source.tex, source lines 6131–6131. Same-chapter pointer into the author environment lemma at source line 6131; invoked at proof line 11928 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EPK: sources/stacks-chow/pinned-original-source.tex, source lines 6167–6167. Same-chapter pointer into the author environment lemma at source line 6167; invoked at proof line 11929 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02TY: sources/stacks-chow/pinned-original-source.tex, source lines 7062–7062. Same-chapter pointer into the author environment lemma at source line 7062; invoked at proof line 11935 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0F9Z: sources/stacks-chow/pinned-original-source.tex, source lines 6184–6184. Same-chapter pointer into the author environment remark at source line 6184; invoked at proof line 11942 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F9J: sources/stacks-chow/pinned-original-source.tex, source lines 10026–10026. Same-chapter pointer into the author environment lemma at source line 10026; invoked at proof line 11968 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FAQ: sources/stacks-chow/pinned-original-source.tex, source lines 10108–10108. Same-chapter pointer into the author environment lemma at source line 10108; invoked at proof line 11971 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02RG: sources/stacks-chow/pinned-original-source.tex, source lines 2401–2401. Same-chapter pointer into the author environment lemma at source line 2401; invoked at proof line 11980 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *The construction above defines a bivariant class*[1]

$$c(Z \to X, \mathcal{N}) \in A^*(Z \to X)^\wedge$$

and moreover the construction is compatible with base change as in Lemma 0FBJ. If $\mathcal{N}$ has constant rank r, then $c(Z \to X, \mathcal{N}) \in A^r(Z \to X)$.

Proof. Since both $i_* \circ j^* \circ C$ and p^* are bivariant classes (see Lemmas 0B78 and 0EPK) we can use the equation

$$i_* \circ j^* \circ C = p^* \circ c(Z \to X, \mathcal{N})$$

(suitably interpreted) to define $c(Z \to X, \mathcal{N})$ as a bivariant class. This works because p^* is always bijective on chow groups by Lemma 02TY.

[1]The notation $A^*(Z \to X)^\wedge$ is discussed in Remark 0FE0. If X is quasi-compact, then $A^*(Z \to X)^\wedge = A^*(Z \to X)$.

Let $X' \to X$, $Z' \to X'$, and $\mathcal{N}'$ be as in Lemma 0FBJ. Write $c = c(Z \to X, \mathcal{N})$ and $c' = c(Z' \to X', \mathcal{N}')$. The second statement of the lemma means that c' is the restriction of c as in Remark 0F9Z. Since we claim this is true for all X'/X locally of finite type, a formal argument shows that it suffices to check that $c' \cap \alpha' = c \cap \alpha'$ for $\alpha' \in \mathrm{CH}_k(X')$. To see this, note that we have a commutative diagram

$$\begin{array}{ccccccc} C_{Z'}X' & \longrightarrow & W'_\infty & \longrightarrow & W' & \longrightarrow & \mathbf{P}^1_{X'} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ C_ZX & \longrightarrow & W_\infty & \longrightarrow & W & \longrightarrow & \mathbf{P}^1_X \end{array}$$

which induces closed immersions:

$$W' \to W \times_{\mathbf{P}^1_X} \mathbf{P}^1_{X'}, \quad W'_\infty \to W_\infty \times_X X', \quad C_{Z'}X' \to C_ZX \times_Z Z'$$

To get $c \cap \alpha'$ we use the class $C \cap \alpha'$ defined using the morphism $W \times_{\mathbf{P}^1_X} \mathbf{P}^1_{X'} \to \mathbf{P}^1_{X'}$ in Lemma 0F9J. To get $c' \cap \alpha'$ on the other hand, we use the class $C' \cap \alpha'$ defined using the morphism $W' \to \mathbf{P}^1_{X'}$. By Lemma 0FAQ the pushforward of $C' \cap \alpha'$ by the closed immersion $W'_\infty \to (W \times_{\mathbf{P}^1_X} \mathbf{P}^1_{X'})_\infty$, is equal to $C \cap \alpha'$. Hence the same is true for the pullbacks to the opens

$$C_{Z'}X' \subset W'_\infty, \quad C_ZX \times_Z Z' \subset (W \times_{\mathbf{P}^1_X} \mathbf{P}^1_{X'})_\infty$$

by Lemma 02RG. Since we have a commutative diagram

$$\begin{array}{ccc} C_{Z'}X' & \longrightarrow & N' \\ \downarrow & & \| \\ C_ZX \times_Z Z' & \longrightarrow & N \times_Z Z' \end{array}$$

these classes pushforward to the same class on N' which proves that we obtain the same element $c \cap \alpha' = c' \cap \alpha'$ in $\mathrm{CH}_*(Z')$. $\square$